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BACHELOR'S THESIS

Software Project

Applied Usage of RPA-Technologies for Improving the Efficiency of HR Process Management in Companies

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Abstract

This paper considers the application of Robotic Process Automation (RPA) technologies for optimization of HR-department processes within a company. The author of the paper determines which HR tasks have processes consisting of routine and repetitive manual work and proposes to speed them up with the help of software bots. Such processes usually do not require intellectual involvement of employees and making any decisions, so their fulfillment can be entrusted to an RPA-bot. The author of this paper has developed and tested three RPA-bots that are able to perform some of the resource-intensive HR tasks in shorter time intervals and with minimal employee involvement. Based on the resulting measurements, we draw a conclusion about the effectiveness and usefulness of using RPA-technology for automating the work of HR department.

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List of Key Words

Robotic Process Automation, Human Resources Management, Business Process Automation, Information technologies, Employee Onboarding.

1. Introduction

Productive work of human resources departments is an integral part of the success of almost any modern enterprise. The main duties of HR specialists are to search and select new qualified personnel, manage the process of evaluating the results of the accomplished work, develop and establish relationships between employees and divisions within the company, develop employee qualification improvement programs, eliminate disputes, and maintain the corporate culture. All of the business processes listed above, among other things, imply the performance of a huge number of various routine tasks of any complexity that are repeated every day in large volumes, which do not require the performer to apply cognitive functions, i.e. decision-making. Such routine operations can take up a significant amount of the total working time of their performers [1].

This report proposes the application of modern robotization process technologies to improve the efficiency of human resource departments of enterprises in various industries. It is planned to analyze different approaches to the automation of human resources processes and tasks, and to compare information technologies currently used by companies. The thesis aims to show the advantages of RPA-technology and efficiency in the use of program robots to accelerate and improve the quality of work of human resources departments. As a result of the thesis, the author is going to develop an RPA-solution consisting of a set of software robots, that implement the automation of part of the tasks of HR employees in order to improve the efficiency of the main business processes of this department.

Concerning the aims and objectives of the project, we are firstly trying to answer the question whether RPA technology can be useful to improve the efficiency of the HR department. To do this, we need to analyze where robotic process automation is applicable in particular, what advantages this technology has, and why it might be useful specifically in this case. After determining which HR processes are eligible for optimization and providing an explanation of why, we should move on to the development of RPA solutions to automate the selected processes. As a part of this, we should demonstrate the efficiency gains that can be achieved by using the RPA software bots that we develop. After completing the development of the RPA solution, a tangible comparison should be made in order to reflect the qualitative result of the completed work.

This paper contains 10 chapters, starting with "Introduction", where we discuss the general idea of the work, state its main aims and objectives.

In the next chapter called "Problem statement" we identify the specific metrics by which we should evaluate the efficiency that we have achieved as a result of implementing our RPA-bots.

In the chapter "Relevance of the topic" we take a brief overview of what exactly HR departments are doing in a company, and propose the application of RPA technology to speed up the HR processes that mainly involve repetitive and time-consuming labor. By demonstrating the presence of such repetitive processes, we explain the importance and relevance of our work and demonstrate that it has a practical application.

In the "Review of existing approaches and methods" we provide a literature review for a number of academic articles, related to the RPA technology, theoretical basis, practical cases of implementation, efficiency and possible risks.

Going further, in the “Selection of the platform” we are explaining our choice of the RPA-platform that we will use in the project. In this chapter we state the main advantages of this software and provide a brief description of its functionality.

In the chapter “The plan for addressing the assigned task”, we describe what HR processes we will optimize in our work: we clarify what these processes consist of, and how the use of robotization could improve the efficiency of their execution. Next, we provide a description and architecture of the database that we will use in our project, with the details for all variables and tables. In this chapter, we also describe the architecture of our bots. For each of them, we outline a step-by-step algorithm that describes how they work and shows what exactly these bots do.

In the chapter “Obtained results” we test our RPA bots on a number of different user queries and demonstrate the achieved results. In addition to that, we compare these results with those of an actual HR employee, who was assigned to perform the same tasks.

Moving forward, in the “Conclusions” chapter we provide the outcomes that we drew from the obtained results. We discuss the insights that we have obtained for each of the three scenarios of robotization, and summarize the overall results of our work.

Going on, in the “Future development” we speculate on the perspectives of the work we have done and suggest in which particular directions this project could be continued. As for the potential perspectives, we describe the possibility of using our work for the implementation of a full-scale automation solution, which could comprehensively cover the HR tasks for organizations in specific industries.

Finally, the “Reference list” contains links to the sources that have been used during the development of this project.

2. Problem statement

We are trying to evaluate the effect that could be achieved by using RPA-bots to perform the routine and repetitive tasks of HR department. Thus, we want to estimate how much we can increase the execution speed of certain processes and how significant this acceleration may be. We are also interested in the possibility of eliminating the human-related mistakes while processes execution; an RPA-bot works according to a predetermined algorithm, unlike a human employee who may make mistakes due to inattention or fatigue. Thus, we are going to justify the utility of the completed work by the increased speed of the process execution and the reduced number of errors made.

3. Relevance of the topic

In order to assess the relevance of this task, we first need to dive into the existing processes and tasks performed by HR in modern companies. The first thing that should be discussed is what HR department is responsible for and what are its main functions and aims. Human Resources is the branch in a company that is responsible for the human resource management — in other words, this department is responsible for maintaining the workforce in the organization. We can identify the main functions of the HR department that form the scope of its responsibility: personnel management, recruitment, employee training and development, evaluation of personnel performance, compensation & benefit (C&B) [2]. Let's make a brief overview of all of them.

For the personnel management, it involves various administrative tasks about employee's performance. This usually includes maintaining employee's data, operating with HR requests, setting up personal schedule and resolving conflicts that appear on the workplace. Going further, recruitment is one of the most important tasks performed by HR: it implies searching for the job candidates that fit company's needs, evaluating their skills and hiring them. This task involves various complicated business processes related to vacancies planning, forming company's requirements, checking resumes and

etc. Moving on to employee training and development, this function is responsible for investing in the development of the company employees' professional skills and competencies, and providing them with the opportunities for further career growth. Monitoring employees' progress in learning new material, estimating training needs, and managing the organization's training processes are just a few tasks where HR is involved in. The next important function of HR is to evaluate the efficiency and quality of staff performance. This task also breaks into several business processes that take a lot of time from HR employees: for example, designing and setting KPI's (Key Performance Indicators) and OKR's (Objectives and Key Results), collecting and working on feedback from company workers, identify those who deserve a bonus and those who underperform, etc. Next, compensation and benefit (C&B) is a function that involves managing employee salary structure, assigning benefits and bonuses. The HR department initiates payments and also monitors the timely fulfilment of paychecks. Also, they should track the market salary levels to maintain competitiveness and optimal wages.

As we can see, a modern HR department handles many different tasks, and each of them consists of a large number of certain business processes that require some amount of effort from HR employees. Moreover, most of the above tasks are time-consuming and resource-demanding for HR staff. If we take, for example, such a typical HR task as reviewing submitted resumes for a job and sending responses, we can notice how long this process could be if it is done manually. Then for the company's management it becomes clear that such sort of tasks as we mentioned above should be optimized: this is the point when modern IT solutions come up [3].

The main focus of this work is the utilization of Robotic Process Automation (RPA) technology and its applicability for the task automation of HR business unit. RPA is a technology that allows to automate the execution of some repetitive and rule-based tasks by using software robots (bots). These bots do repetitive and routine work,

replacing human employees, and thus saving their time and resources while speeding up the entire working process, following the compliance and reducing the number of errors. RPA bots operate with other software through the API or directly through the user interface. This technology has received a lot of attention in the last decade and many large companies are implementing RPA solutions in order to optimize the performance of their various departmental processes. Currently there are many RPA-vendors that are leading on the international market; for example, we can mention UiPath, Automation Anywhere, Blue Prism, Microsoft Power Automate and etc. On the Russian market, one of the most recognizable local solutions is Pix RPA platform; subsequently, we will implement our research aims using this particular software and later the choice of software will be explained.

4. Review of existing approaches and methods

One of the most interesting papers on this topic is 'Improving Efficiency and Effectiveness of Robotic Process Automation in Human Resource Management'. It describes the development of a human resource management system (HRMS) model using RPA-bots to optimize repetitive manual tasks. In this paper, the authors show a step-by-step methodology for developing an RPA solution, from process selection and preparation to testing accuracy and speed of execution. As for results, the authors developed a number of RPA bots that optimize some HR-tasks related to operation with employee's personal data. These bots are designed for gathering, storing and accessing data from the database and performing some repetitive tasks using this data. In the conclusion of the paper, the authors emphasize the reliability of RPA bots in terms of accomplishing their tasks in comparison to real employees. The reliability and efficiency are especially noticeable for large amounts of work; the authors highlight the accuracy of bots in comparison to real employees, who tend to make mistakes when performing large tasks due to fatigue or poor attention. As a result, the authors succeeded in developing an RPA solution that significantly accelerated the execution of HR processes and greatly reduced the amount of workload carried out by HR employees [4].

The article "The Future Digital Work Force: Robotic Process Automation (RPA)" explores RPA technology as a solution that can radically change the approach to automation of routine manual tasks. This paper explores the knowledge in this direction in details, giving possible implementation scenarios of RPA technology and illustrating them with real business cases of large companies. The authors of the paper give a brief overview of possible applications of RPA technology. In particular, they emphasize the possibility of using software bots in the IT sphere, telecom, financial services and banking, insurance industry, healthcare, manufacturing, and energy sector. The authors

note that in such different sectors we can find business processes that are quite similar in their nature and problematics, so they can be optimized by using software bots [5].

The article "Automating HR processes with Robotic Process Automation (RPA)" is a qualitative review of the application of RPA technology for optimizing HR processes with a focus on practical application. The article describes in details in which processes of HR department automation with the help of software bots can achieve maximum efficiency. The overview of this approach also covers its application in the financial sector, as well as integration with modern artificial intelligence systems and the possibility of using machine learning algorithms for other projects of a larger scale. The author concludes that with the use of RPA-technologies there will be a transformation of the roles of HR-employees in companies: they will have more time to focus on such things as talent management, organizational development and culture building, while processes that do not require active intellectual work could be performed by software bots [6].

The article "Robotic Process Automation (RPA) as a Digitalization Related Tool To Process Enhancement and Time Saving" analyzes RPA technology as a digital transformation tool for improving process efficiency. Special attention in the article is paid to business cases that show the effectiveness of this technology in different business areas. As for examples, it mentions the use of software bots for data management, optimization of document preparation process and automation of financial calculation processes. A significant part of the paper is focused on SWOT-analysis for RPA-technology. The author has conducted a comprehensive study of the strengths and weaknesses of this technology, as well as identified potential opportunities and threats. As strengths, the author notes significant cost reduction, increased business efficiency and quality of work, as well as employee satisfaction and faster job completion. As weaknesses of RPA-technology the author pointed out the need to intervene in the work of program bots in case of errors, as well as the need for

high quality data. Potential opportunities of RPA-technology are integration with AI, solving problems with staff shortages and transformation of employee roles in favor of more intellectual work. Possible threats are the emergence of specialized software solutions that make the development of RPA-bots less useful, problems with change management in companies and development problems. As a conclusion of this work, the author emphasizes that the use of RPA can bring real economic and operational benefits to a company if the processes to be automated are properly chosen and the potential threats are taken into account [7].

The article "Risks Associated With Robotic Process Automation" analyzes the potential risks that emerge during the application of RPA-technology. The authors have performed a detailed review of other works and business cases on the topic of software robotization, as a result of which several classifications of possible risks associated with the RPA implementation were presented. The authors focus on several key risks that affect the overall success from the use of software bots; these risks include technology risks, human factors, financial risks, and the risks of making the incorrect choices of business processes to automate. Technology risks include software compatibility issues, security issues, and vulnerabilities due to changes in interface design (especially for bots that are interacting with web forms). The human factor involves change management problems, such as resistance to change from employees and lack of staff competence in dealing with new technology. Regarding financial risks, the authors mention that expectations of cost reductions are often overstated. Furthermore, they mention the waste of resources in the development of an obviously unsustainable automation project. Among other things, the authors highlight the risks of choosing unsuitable processes for automation; often customers do not understand what RPA technology is used for and to which processes it can be applied, which leads to costly and inefficient projects. The authors conclude their work with the opinion that the adoption of RPA technology requires not only an understanding of the technical aspects,

but also business knowledge and a proper assessment of all potential risks for each particular case [8].

5. Selection of the platform

As previously mentioned, the RPA-platform Pix RPA was chosen to implement the stated objectives. One of the key advantages of this platform is its flexible and convenient interface, which allows not only to write your own scripts in C#, but also to take advantage of low-code systems, in particular, to use the built-in activities. Having such a variety of standard activities, as well as documentation for each of them, makes the development process much simpler. Thus, due to the combination of the advantages of the low-code platform and the C# language, we can develop our scripts significantly faster. Another remarkable advantage is flexibility and scalability. This platform provides all the tools essential for working with attended and unattended bots (attended bots – run in user environment, unattended bots – run mostly on the server side without user interaction); in addition, there are also necessary activities for the integration with the web browsers and operating system elements, as well as various assets for integration with databases engines (MS SQL Server, Oracle, PostgreSQL, MySQL, etc.), messengers (WhatsApp, Telegram, etc.), office application packages, and so on. In addition, this vendor of this software is a Russian company, which minimizes the possible problems with licensing and allows to get technical support and maintenance much faster.

5.1 Chapter conclusions

As a result of this chapter, we decided to select Pix RPA platform for the implementation of our project. This particular software fits all our requirements, including the integration with MSSQL databases engine and Microsoft Office. In addition to that, it is possible to use an add-on activity for setting up the integration with Telegram messenger: this is especially beneficial if we want to organize the data input for one of our bots in a dialog format.

6. The plan for addressing the assigned task

We also should outline where exactly the RPA technology can be applied. As stated earlier, this technology includes the creation of software bots that can emulate the work of a human employee in routine and repetitive tasks. Thus, we can suggest the following types of tasks that can be handled by this technology: routine (repetitive, identical) tasks with lots of iterations, structured rule-based tasks, and tasks involving dealing with large amounts of data, as well as on-demand manual tasks. In addition, RPA technology is extremely effective in building integration with legacy or custom-developed systems (usually, such systems do not have integration points like API or web-services) [9].

We need to conduct an analysis of HR tasks that would show us for which specific processes we could apply RPA technology. From our list, the most appropriate process for robotization is the process of recruiting new staff. The process of hiring new employees necessarily involves a search for candidates. Often, the search for specialists is being performed through special job portals, where candidates post their CVs and express their desire to find a job in a particular area. The candidate search process includes the selection of profiles that meet the needs and policies of the company. This process fits perfectly with the description of the process for robotization that we outlined above; it is a routine, time-consuming activity that involves gathering information about potential job candidates and saving that information for later processing and communication with them. Thus, we can develop a software bot that searches for the specific profiles for the required parameters and then saves the obtained data into an Excel document or a database. We will develop our solution based on priorities on the Russian market. In Russia, one of the most popular employee search portals is HeadHunter's “hh.ru” service. Thus, we have to develop a bot that interacts with the Web-interface of the “hh.ru” portal and emulates the work of an HR employee.

Moving on from this, Compensation and Benefits (C&B) HR unit also has suitable processes for robotization. This field of activity of the HR department involves working with data and calculating certain indicators. Among these processes, there are the ones that are repeated at certain intervals, calculating values according to some certain pattern or template. One of the examples of such processes can be the calculation of employee salaries in a form of “Base salary plus variable commission”. This type of salary implies that an employee has a fixed monthly salary, as well as bonuses or commissions for the completed deals and sales made at the end of every month. Often this form of salary is applied to employees of sales department, where the bonus for each sale is calculated according to some coefficient (not necessarily constant for each deal). This process perfectly fits the criteria that we have established earlier; in this case, the human work could be replaced by a bot that accesses the company's database, looks up the necessary information for each employee, and calculates their full monthly salary (the sum of salary and commission). This bot can save a considerable amount of time that an ordinary HR employee usually spends on searching the database and calculating salaries manually.

Another activity that we want to automate is personnel management. As we have described earlier, the responsibilities of HR employees include the preparation of various documents requested by company employees. One of such documents is a certificate of employment. Such certificates are often made according to the same template, they contain personal information about the employee and his/her position in the company. The preparation of such documents and their equivalents is often a routine activity that HR department employees perform upon the requests of the company staff. This process is an on-demand manual process, which involves extracting the necessary information about the employee and drafting a document based on a template. Such work could also be replaced by some kind of an RPA-bot that would access the company's employee directory (a database, Active Directory, ERP system, etc.) and

then prepare a compiled document of the specified format (usually, Microsoft Excel, Word or .pdf document). This is another good example of a process where the usage of RPA technology impacts the company a lot, with greater efficiency. Implementing the interaction with an RPA-bot via a popular messenger (for example, Telegram), we can reduce the total time spent by a company employee for communication with HR department, thereby freeing the latter from the time-consuming task, that they can spend for more intellectual work.

It is also worth mentioning the other HR tasks we described at the beginning of our study. When we are trying to automate certain processes, we should always keep in mind for what exactly our approaches are used. In our case, we first outlined where exactly RPA technology is applicable, and then listed the processes in the HR department where this technology can give a significant increase in productivity. Regarding employee training and development, this process mostly requires a personalized approach to each employee. This means that in order to effectively perform these duties, an individualized approach to each particular case is essential. This type of tasks does not contain repetitive and identical elements, but on the contrary, require a lot of intellectual resources. Thus, in this type of HR tasks robotization is not applicable by its definition. Moving on to the evaluation of personnel performance, here the question is raised about the advisability of using the RPA-technology for this. Unlike all previous HR tasks, there are plenty of software solutions for performance evaluation that allow managers to calculate the necessary coefficients quickly. This is a whole separate specialty that has its own software and its own unique methods of evaluation and visualization of outcomes. RPA technology focuses more on automating certain actions within business processes and speeding up manual work. Thus, RPA-technology will not be the most effective solution in this area, as it does not fit the main criteria in which this type of automation is used.

To store the data of this project securely and performing data management, we will need to use a database. We will use MS SQL server, which will have an architecture that simulates the database of a real organization. We will use synthetic data, as the purpose of our project is first of all to demonstrate the effectiveness of RPA-technology application in the field of HR.

Below you can see a visualization of the database architecture we will use in our work, as well as comments on each table and the descriptions for the variables [Fig.1].

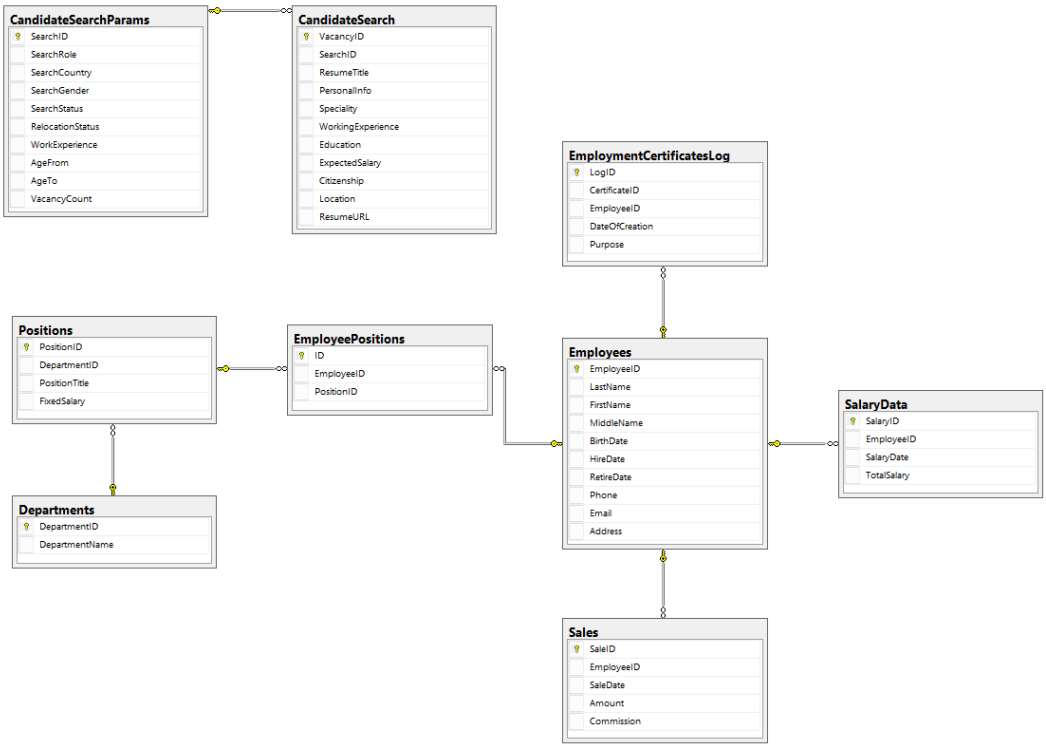


Fig. 1. Database diagram of the project

6.1 Database description

Employees – this table contains information about company's staff. It includes employee's full name, contact data and dates of birthday, hiring and retirement (last one for the case if some employee had left the company). Description is presented on Table 1.

Table 1. Employees table

Column name	Datatype	Description
EmployeeID	int	ID of an employee (primary key, autoincremented)
LastName	nvarchar(50)	Last name of an employee
FirstName	nvarchar(50)	First name of an employee
MiddleName	nvarchar(50)	Middle name of an employee
BirthDate	date	Birth date of an employee
HireDate	date	Day when employee was hired
RetireDate	date	Day when employee retired and left the position (empty for employees who are not retired)
Phone	nvarchar(50)	Phone number of an employee
Email	nvarchar(50)	Active email of an employee
Address	nvarchar(255)	Address of an employee

Sales – this table contains a historic data about successful sales made. Each sale is supervised by one of the company's employees, so his/her name is saved, together with the monetary amount of sale and the commission percentage, assigned to the salesman. Description is presented on Table 2.

Table 2. Sales table

Column name	Datatype	Description
SaleID	int	ID of the sale (primary key, autoincremented)
EmployeeID	int	ID of an employee who is responsible for this sale (foreign key)
SaleDate	datetime	Date when sale happened
Amount	decimal(18, 2)	Sale amount
Commission	decimal(18, 2)	Commission percentage for the salesman

SalaryData – this table contains a historic data about payrolls. For each particular payroll, it contains the ID of a recipient, date when payroll was made and the total amount paid. Description is presented on Table 3.

Table 3. SalaryData table

Column name	Datatype	Description
SalaryID	int	ID of the payroll (primary key, autoincremented)
EmployeeID	int	ID of an employee who receives this salary (foreign key)
SalaryDate	datetime	Date when the salary was paid
TotalSalary	decimal(18, 2)	Total amount of salary

EmploymentCertificatesLog – this table contains a historic data about employment certificates that were created. For each log, there is an ID of a certificate, ID of an employee who ordered it, date of creation and an explanation of the purpose for creating this certificate (in text form). Description is presented on Table 4.

Table 4. EmploymentCertificatesLog table

Column name	Datatype	Description
LogID	int	ID of the log (primary key, autoincremented)
CertificateID	int	ID of the certificate
EmployeeID	int	ID of the employee who ordered the certificate (foreign key)
DateOfCreation	datetime	Date when the certificate was created
Purpose	nvarchar(255)	Purpose for giving this certificate

EmployeePositions – this table is a technical one, it was created in order to simplify Many-To-Many relation between tables Employees and Positions. For each relation, it contains ID of employee and an ID of this employee's position. Description is presented on Table 5.

Table 5. EmployeePositions table

Column name	Datatype	Description
ID	int	ID of connection (primary key, autoincremented)
EmployeeID	int	ID of the employee who occupies a certain position (foreign key)
PositionID	int	ID of the position taken by a particular person (foreign key)

Positions – this table contains information about all positions that exist in a company. For each position, there is an ID of the department in belongs to, its full title and fixed salary. Description is presented on Table 6.

Table 6. Positions table

Column name	Datatype	Description
PositionID	int	ID of a certain position in the company (primary key, autoincremented)
DepartmentID	int	ID of the department to which the position belongs to (foreign key)

PositionTitle	nvarchar(255)	Name of the position
FixedSalary	decimal(18, 2)	Fixed amount of salary assigned to this position

Departments – this table contains information about the departments in the company. For each ID in this table, there is a full title of the department. Description is presented on Table 7.

Table 7. Departments table

Column name	Datatype	Description
DepartmentID	int	ID of the department (primary key, autoincremented)
DepartmentName	nvarchar(255)	Name of the department

CandidateSearchParams - this table contains the ID of the launch, as well as the selected parameters that user has specified. Description is presented on Table 8.

Table 8. CandidateSearchParams table

Column name	Datatype	Description
SearchID	int	ID of bot launching (primary key, autoincremented)
SearchRole	nvarchar(500)	Specialization of the employee the company is looking for
SearchCountry	nvarchar(200)	Citizenship of the candidate
SearchGender	nvarchar(200)	Gender of the candidate
SearchStatus	nvarchar(200)	Job search status given by the candidate
RelocationStatus	nvarchar(200)	Candidate's status regarding relocation
WorkExperience	nvarchar(200)	Candidate’s work experience
AgeFrom	nvarchar(200)	The minimum age of the candidate
AgeTo	nvarchar(200)	The maximum age of the candidate
VacancyCount	nvarchar(200)	Number of profiles searched during the launching

CandidateSearch – this table contains data on searched profiles, which were obtained by using our bot. The primary key is VacancyID, which is the identifier of the found profile; SearchID contains the bot launch identifier, during which this profile was found. This table includes the title of the profile and brief personal information about the candidate, as well as his/her specialization, work experience, education, expected

salary, citizenship, place of residence and a link to this profile on “hh.ru”. Description is presented on Table 9.

Table 9. CandidateSearch

Column name	Datatype	Description
VacancyID	int	ID of the profile (primary key, autoincremented)
SearchID	int	ID of the search in which the profile was found (foreign key)
ResumeTitle	nvarchar(255)	Title of the profile
PersonalInfo	nvarchar(MAX)	Brief information about the candidate
Speciality	nvarchar(1000)	Candidate specialization
WorkingExperience	nvarchar(MAX)	Working experience of the candidate
Education	nvarchar(MAX)	Brief information about the candidate's education
ExpectedSalary	nvarchar(255)	Candidate’s expected salary (if provided)
Citizenship	nvarchar(255)	Candidate’s citizenship
Location	nvarchar(500)	Candidate’s place of living
ResumeURL	nvarchar(500)	URL to the original profile

6.2 RPA-bots description

6.2.1 Description of the bot for profile searching

- The bot starts with generating an ID of the launch, and then allows the user to set up the parameters in the dialog window for searching the required profiles [Fig.2]. After entering the parameters and the number of vacancies, the user clicks “OK” button, and then the bot performs all the searching job on its own.
- Bot starts its search by launching the Google Chrome browser and navigating to the “hh.ru” portal via the URL address. This address was previously constructed according to the parameters that user entered in the form.
- Bot loads the page where the required profiles are listed one by one [Fig.3]. In this representation the majority of required data is listed on the page, so our bot starts parsing. Initially, bot scans all the data using the XPath, and stores it in temporary files; then it processes the data for each particular profile, removing the unnecessary constructions and saving the obtained data in the resulting Excel document. During the process described above, the bot obtains the information

about the resume title, specialization, working experience, education, citizenship, current location and link to the original resume. Candidate's brief description and expected salary are listed inside each of these profiles, so our bot needs to iterate through all of them and collect this data.

- Bot navigates the profile links one by one and collects data about the candidate and the preferred salary [Fig. 4]. On this stage, bot has prepared an Excel document that was requested by user, with styles and all necessary information for the candidates [Fig. 5].
- The last step is saving all the data to the company's database. For this, bot uses the Pix RPA platform's built-in activity for interaction with databases. The bot creates an SQL query and saves the resulting table to the database, additionally putting the ID of the search in a separate column.
- After performing all the above operations, the bot closes the Chrome browser.

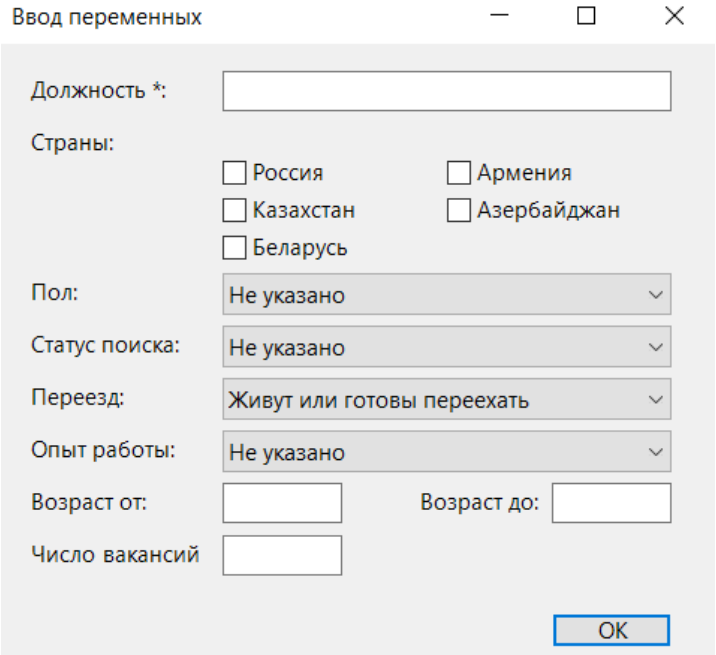


Fig.2. Input form

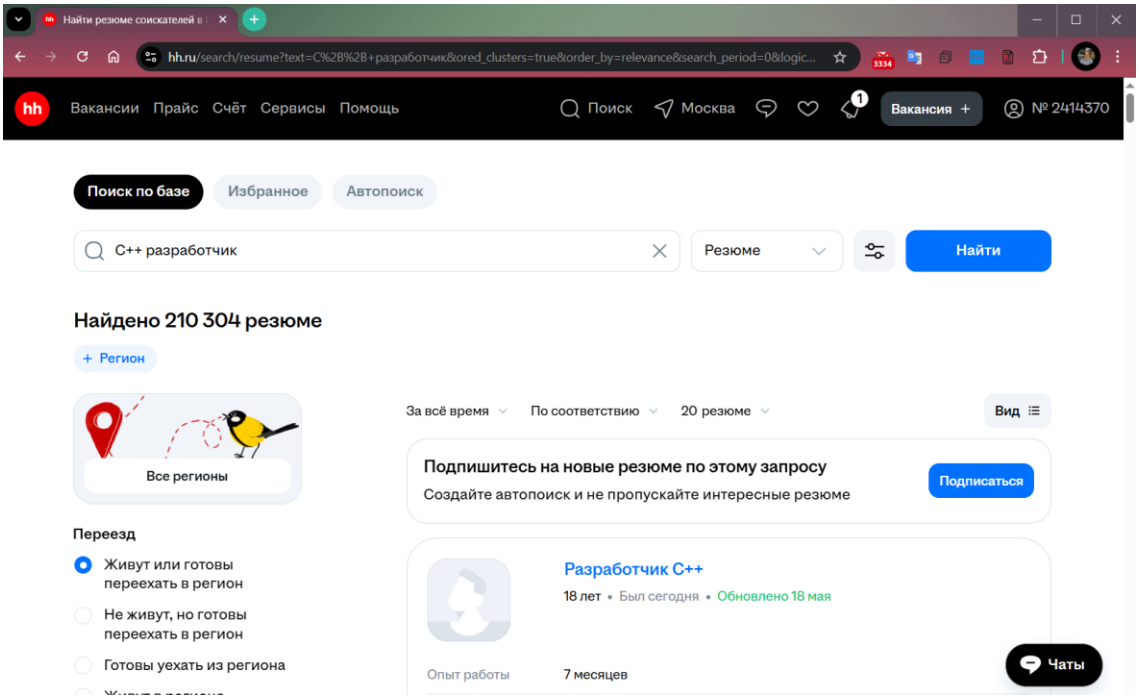


Fig.3 "hh.ru" portal page, bot navigates it after obtaining the search parameters

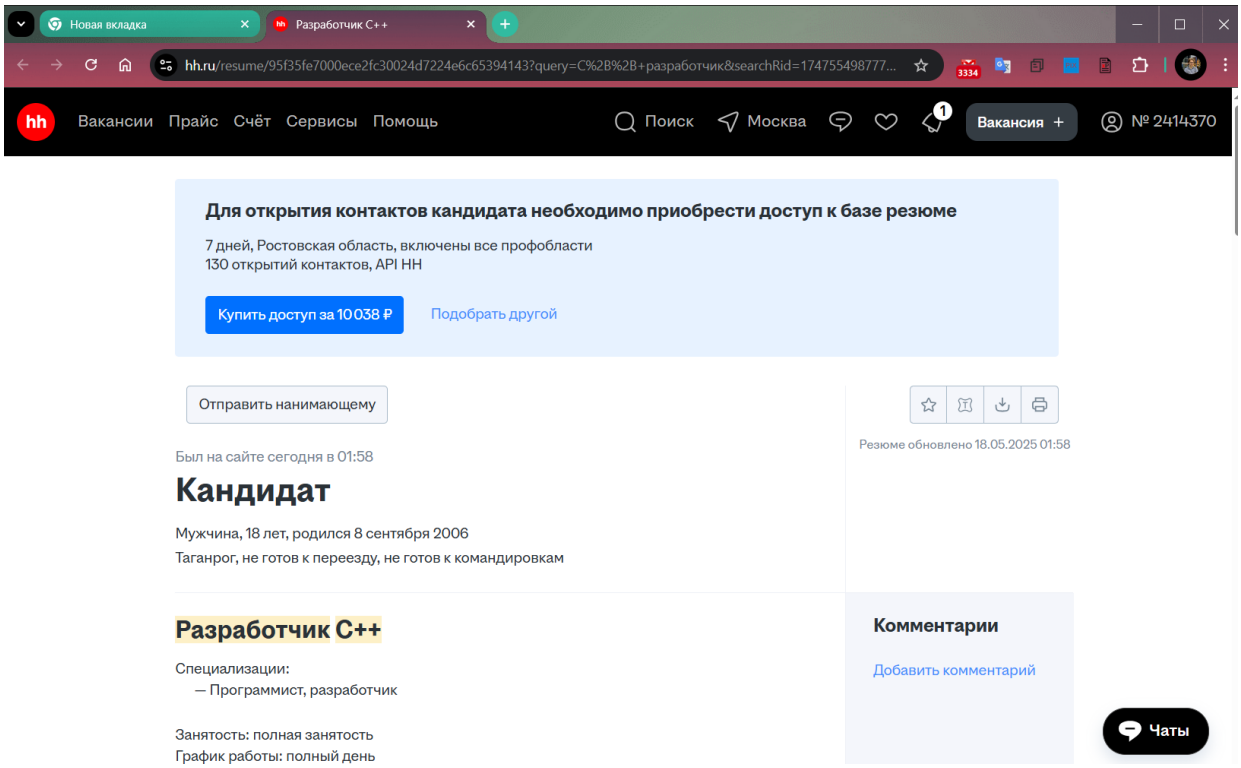


Fig.4 Interface of the profile page

	A	B	C	D	E	F	G	H	I
1	Resume title	Brief personal information	Speciality	Working experience	Education	Expected salary	Citizenship	Current location	Link to the resume
2	Разработчик C++	Мужчина, 18 лет, родился 8 сентября 2006	Программист разработчик	7 месяцев	Колледж цифровых технологий TOP IT • 2026 Знание языков Родной язык — Русский		Россия	Таганрог • Переезд невозможен	https://hh.ru/resume/95f
3	Разработчик C++	Мужчина, 24 года, родился 10 июня 2000	Программист разработчик	4 года 5 месяцев	Уфимский университет науки и технологий Уфа • 2023 Знание языков Родной язык —		Россия	Уфа • Переезд невозможен	https://hh.ru/resume/aaf
4	Разработчик C++	Мужчина, 23 года, родился 27 февраля 2002	Программист разработчик	1 год 8 месяцев	Санкт-Петербургский государственный морской технический университет Санкт-Петербург • 2024 Знание языков Родной язык —		Россия	Санкт-Петербург • Переезд невозможен	https://hh.ru/resume/89t
5	Разработчик C++	Мужчина, 25 лет, родился 15 декабря 1999	Программист разработчик	4 года 10 месяцев	Не указано		Россия	Москва • Переезд невозможен Знание языков Родной язык — Русский Английский B1 — Средний	https://hh.ru/resume/ff7
6	Разработчик C++	Мужчина, 20 лет, родился 13 августа 2004	Программист разработчик	2 года 1 месяц	Не указано		Россия	Шимановск • Возможен переезд Знание языков Родной язык — Русский Другие резюме Ещё 1 резюме	https://hh.ru/resume/f8e
7	Разработчик C++	Мужчина, 18 лет, родился 16 марта 2007	Программист разработчик	1 год 10 месяцев	Не указано	70 000 ₽ на руки	Россия	Санкт-Петербург • Переезд невозможен Знание языков Родной язык — Русский Английский B1 — Средний Другие резюме Ещё 1 резюме	https://hh.ru/resume/93

Fig.5 Resulting Excel table with collected profiles

The profile data that was saved to the company's database could be used on the further stages of approval by HR employees, so we decided to demonstrate the usability of our RPA-bot by adding to our project a web-interface for interacting with collected data [Fig. 6]. The interface that we have developed could be used for finding vacancies among those ones that were collected by our bot. The web form was done with K2 Low-Code Designer, which makes it very easy to develop the interface and set up the communication with the database without any coding.

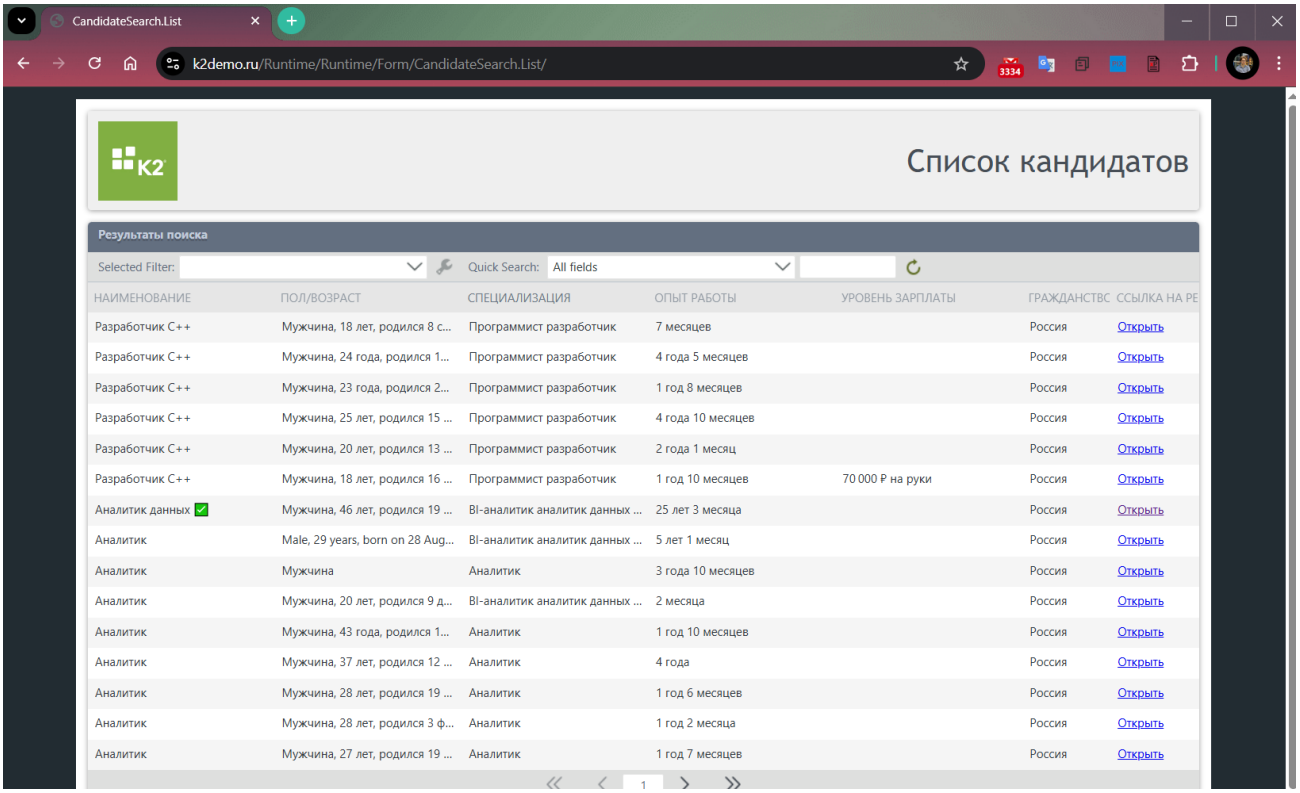


Fig. 6 Web-interface for interacting with collected data

6.2.2 Description of the bot for calculating salaries

- The bot starts with opening a form for initial details input by the user [Fig.7]. On this form a user can select some particular year and month, for which he can calculate salaries of the employees. After the form is completed, the user clicks “OK” to initiate the calculation process.
- Bot starts execution. It makes a request to the company’s MSSQL server. To calculate salaries and outputting the amount for the month, bot needs to find out the performed sales activities of company’s employees, get their fixed salaries corresponding to their roles and calculate the total commission and total salary.
- The resulting data is saved to the Excel spreadsheet which is being generated from the template and may contain some corporate styling [Fig. 8].
- The bot sets up a correct data format for the columns (monetary values in Russian rubles).

- The bot saves the file and stops execution.

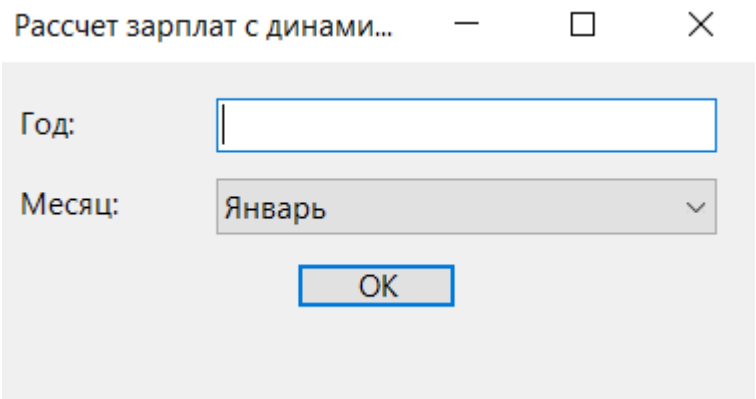


Fig. 7 Form for input data from the user

	A	B	C	D	E	F
1	EmployeeID	PositionTitle	SaleCount	FixedSalary	CommissionValue	TotalSalary
2	48	Экономист	1	129,528.56 ₽	1,995.63 ₽	131,524.19 ₽
3	100	Экономист	1	129,528.56 ₽	767.95 ₽	130,296.51 ₽
4	56	SEO-специалист	1	117,567.81 ₽	3,378.38 ₽	120,946.19 ₽
5	55	SEO-специалист	1	117,567.81 ₽	574.78 ₽	118,142.59 ₽
6	20	SEO-специалист	1	117,567.81 ₽	345.92 ₽	117,913.73 ₽
7	19	Менеджер по закупі	4	99,282.26 ₽	2,886.52 ₽	102,168.78 ₽
8	57	Логист	3	80,342.38 ₽	4,148.09 ₽	84,490.47 ₽
9	51	Логист	1	80,342.38 ₽	3,348.08 ₽	83,690.46 ₽
10	63	Инженер	3	63,344.56 ₽	3,361.68 ₽	66,706.24 ₽
11	95	Помощник руководі	1	49,518.49 ₽	314.09 ₽	49,832.58 ₽
12	38	Рекрутер	2	44,979.14 ₽	560.91 ₽	45,540.05 ₽
13	93	Бухгалтер	1	40,432.66 ₽	792.36 ₽	41,225.02 ₽
14	92	UX/UI-дизайнер	1	35,498.93 ₽	3,332.51 ₽	38,831.44 ₽
15						
16						

Fig.8 Salaries that were calculated by the bot and presented in an Excel format

6.2.3 Description of the bot for generating employment certificates

This bot will perform communication with a user via Telegram messenger, and generate the employment certificate by addressing company's database. We decided to organize the architecture of this bot in the following way: we have created a master RPA-script that is responsible for integration with Telegram messenger and dialog with users and a sub-bot responsible for generating certificates that can be triggered by the master. The sub-bot obtains the parameters that user imputed via chat, gets all the necessary information from a company's database and generates an employment certificate.

For the master bot we have used an activity called “Telegram Manager” that was introduced by Pix Robotics company. We created a chat for our bot by using the special Telegram services, and used a generated security token to connect to it.

- To start a dialog with our bot, a user needs to write a greeting word to the chat; the bot will respond with brief instructions on how to enter the required data properly [Fig. 9]. Each step of communication is denoted by some state value, which is stored in an integer variable.
- When the dialog with the user progresses to the next stage, our bot goes to the next state and executes its commands [Fig. 10]. If an error occurred on some of the states due to the wrong format of inputted data, bot will inform the user and ask to re-apply the application for employment certificate. Bot saves states for each user independently (using dictionary): this means that bot could be used by several people at one time.
- For the script that generates the employment certificates, it starts with establishing a connection to the database and creates a request to the MSSQL server of the company. It is necessary for obtaining employee’s position, ID, date of hiring and size of salary; this data will be used to generate the required document.
- Then script fills the document according to the predefined template given by the company, inserts all the necessary data and obtains the ID of the resulting employment certificate.
- The bot saves the resulting document to the predefined folder, where it can be picked up by the master script that is responsible for the communication with Telegram.
- After this, the generated document is sent to the user via the messenger [Fig. 11].

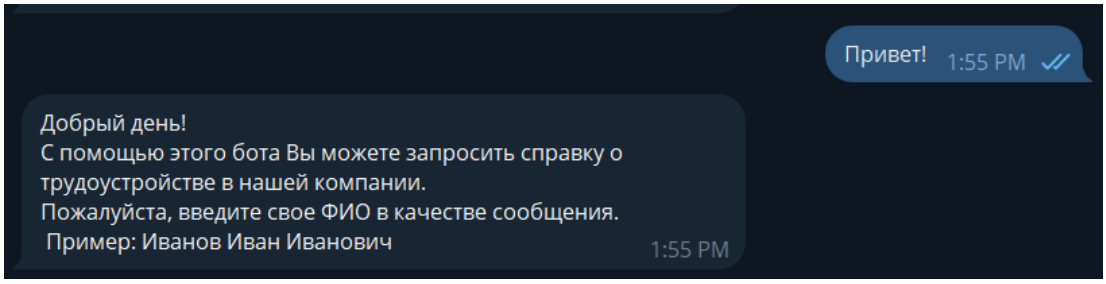


Fig.9 User initiates dialog with the bot

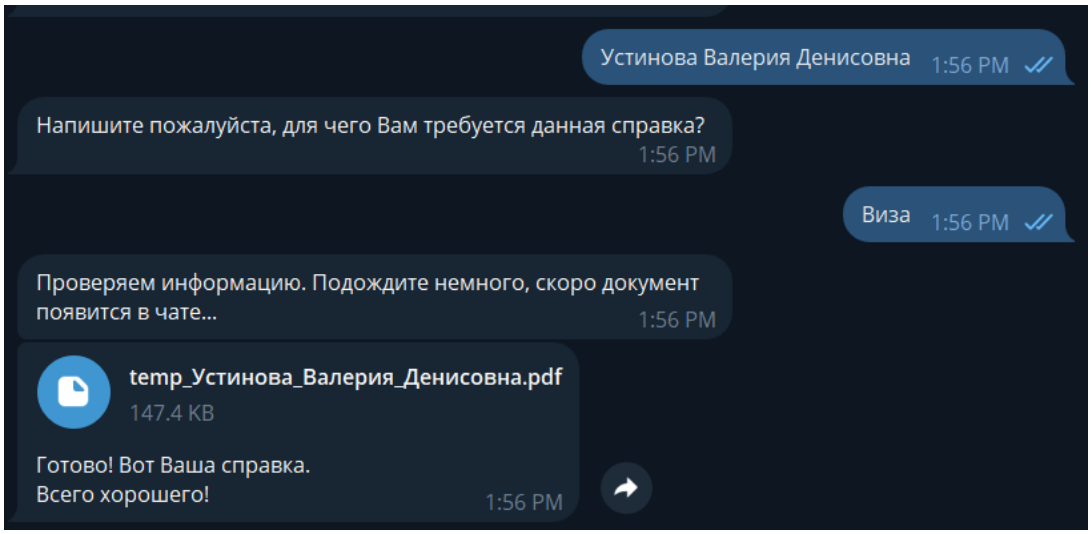
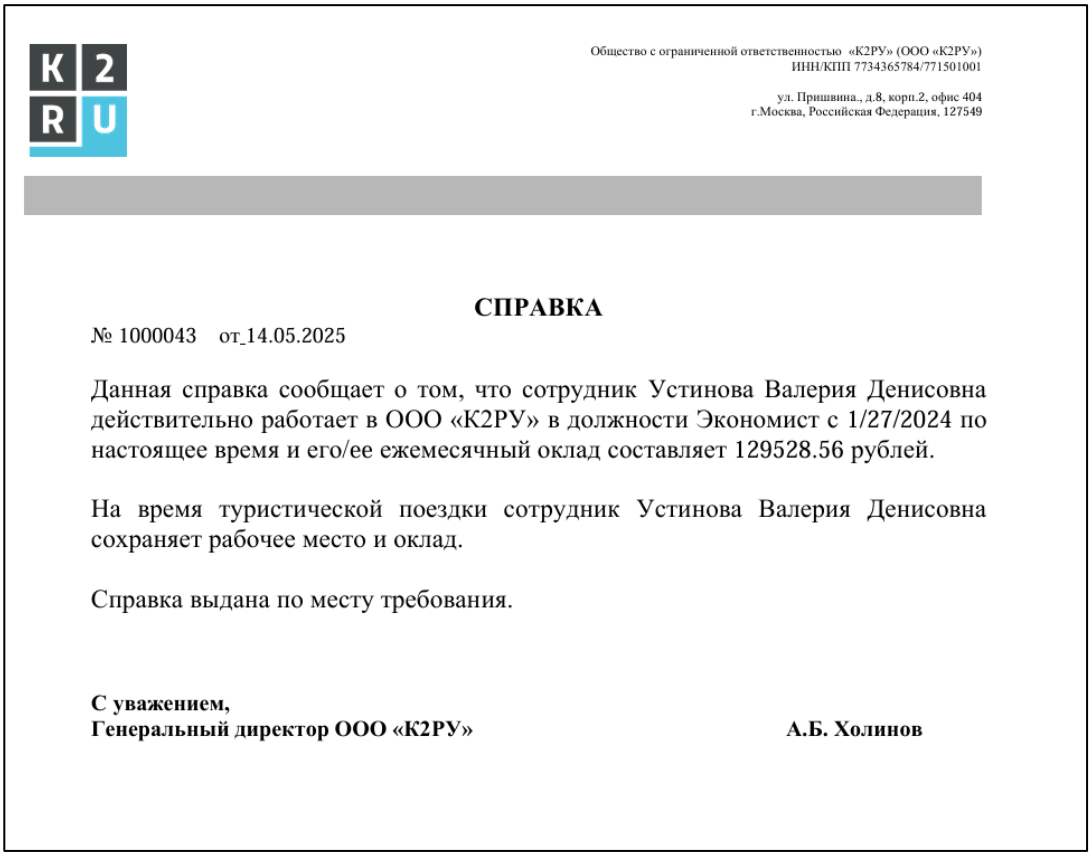


Fig.10 Interaction with the bot via the messenger



Общество с ограниченной ответственностью «К2РУ» (ООО «К2РУ»)
ИНН/КПП 7734365784/771501001

ул. Пришвина., д.8, корп.2, офис 404
г.Москва, Российская Федерация, 127549

СПРАВКА

№ 1000043 от_14.05.2025

Данная справка сообщает о том, что сотрудник Устинова Валерия Денисовна действительно работает в ООО «К2РУ» в должности Экономист с 1/27/2024 по настоящее время и его/ее ежемесячный оклад составляет 129528.56 рублей.

На время туристической поездки сотрудник Устинова Валерия Денисовна сохраняет рабочее место и оклад.

Справка выдана по месту требования.

С уважением,
Генеральный директор ООО «К2РУ» **А.Б. Холинов**

Fig.11 Example of the generated employment certificates for one of the employees of the company. The screenshot is taken from the PDF document that was generated by the RPA-bot

6.3 Chapter conclusions

In this chapter, we have outlined the specific HR tasks that are addressed by our RPA-bots. Additionally, we have given a detailed structure of our project. For each of the selected categories of tasks that we have chosen as suitable for robotization, we have picked the specific HR duties that we will work with, and for which we aim to reach the maximum efficiency. Within the separate paragraphs, we have described the database architecture that we used in our project, as well as provided the descriptions of how each of our RPA-bots is operating. We have categorized the operation process of each of the bots into the separate points, and demonstrated the way each of our RPA-scripts works. As a result of this chapter, we have developed the three RPA-bot according to the architecture that we have described in these paragraphs.

Links to the RPA-scripts are published on GitHub, and could be accessed by the link that is stated in the Appendix [11]. URL of the web-form for reviewing results of profile searching is stated in Appendix [12].

7. Obtained results

In order to demonstrate the effectiveness of our bots, we should test their functionality, record their time of execution and evaluate the result. For our case, the most representative comparison is to compare the results of our test with the real activities of human employees. In order to perform this task, I decided to collect the data from an HR-specialist in the company where I work. The HR-employee should perform the same tasks with the same input parameters as the bot will do. We will focus our attention mostly on the execution time; also, we will validate the obtained result for errors and ensure the quality of the results.

Analyzing the performance of the first bot, I decided to do the following. I decided to design nine different candidate requests; they are presented in the Table 10 below.

Table 10. Requests for the first bot

Request number	Position	Citizenship	Gender	Search status	Relocation	Work experience	Age: lower bound	Age: Upper bound	Number of profiles
Request 1	ML-Engineer	Russia	Female	Looks for offers	Live in their place or ready to relocate	1-3 years	20	-	4
Request 2	Frontend-developer	Russia	Male	Looks for offers	Live in their place or ready to relocate	3-6 years	25	50	5
Request 3	.NET developer	Russia	Male	Actively searches for the job	Live in their region	1-3 years	20	40	6
Request 4	System Analyst	Russia, Kazakhstan, Belarus, Armenia, Azerbaijan	Any	Looks for offers	Live in their region	-	30	45	8
Request 5	Web-developer	Russia, Kazakhstan, Belarus	Any	Actively searches for the job	Live in their place or ready to relocate	1-3 years	25	-	10

Request 6	System Administrator	Russia	Male	-	Live in their place or ready to relocate	3-6 years	-	50	12
Request 7	Data-Engineer	Russia	Male	Looks for offers	Not living in the region, ready to relocate	-	21	49	14
Request 8	DevOps-Engineer	Russia	Male	Looks for offers	Live in their place or ready to relocate	-	25	49	16
Request 9	Tester	Russia	Male	Looks for offers	Live in their place or ready to relocate	-	-	40	18

These requests will be offered to both the real employee and the bot I have created. In the Table 11 below you can find the resulting outcomes.

Table 11. First scenario results

	Request 1	Request 2	Request 3	Request 4	Request 5	Request 6	Request 7	Request 8	Request 9
Human HR-employee	12 min 44 sec	14 min 58 sec	17 min 34 sec	21 min 18 sec	24 min 14 sec	26 min 34 sec	27 min 21 sec	28 min 5 sec	28 min 44 sec
RPA-bot	1 min 10 sec	1 min 17 sec	1 min 30 sec	1 min 56 sec	2 min 36 sec	2 min 53 sec	2 min 58 sec	3 min 46 sec	4 min 2 sec

For the second bot, the task is to make a calculation of employee salaries for five months randomly selected by me. These months are indicated in the Table 12 below.

Table 12. Requests for the second bot

Request name	Year	Month
Request 1	2024	May
Request 2	2024	August
Request 3	2024	November
Request 4	2025	January
Request 5	2025	March

For comparison, a real employee is given the exact same task with the same dates. The results by time are presented in the Table 13 below.

Table 13. Results for the second scenario

	Request 1	Request 2	Request 3	Request 4	Request 5
Human HR-employee	11min 43 sec	12 min 34 sec	11 min 22 sec	10 min 56 sec	11 min 15 sec
RPA-bot	9 sec	8 sec	9 sec	10 sec	9 sec

The task of the third bot is to generate certificates to 18 different employees of the company. Relevant employees will be selected and the reasons why they take the certificate will be provided. The input parameters for testing can be found in the Table 14 below.

Table 14. Requests for the third bot

Request number	Name of the employee	Reason
Request 1	Устинова Валерия Денисовна	“Хочу получить визу”
Request 2	Вишняков Леонид Вячеславович	“Получение визы”
Request 3	Филиппова Василиса Всеволодовна	“Беру для новой визы”
Request 4	Кириллова Александра Вениаминовна	“Нужно для оформления документов”
Request 5	Евсеев Виктор Евдокимович	“Хочу оформить справку для отпуска.”
Request 6	Полякова Лидия Арсениевна	“Для съема квартиры”
Request 7	Логинов Ярополк Гуриевич	“Нужно подать документы на визу. Нужна справка о трудоустройстве”
Request 8	Гордеев Зиновий Олимпиевич	“Оформляю документы на визу”
Request 9	Якушева Евдокия Гостомысловна	“Для съема жилья”
Request 10	Юдина Элеонора Творимировна	“Нужно подтвердить занятость”
Request 11	Пахомов Лавр Проклович	“Оформление визы”
Request 12	Громова Иванна Симоновна	“Нужна справка для получения визы”
Request 13	Гришина Елизавета Егоровна	“Нужна справка для съема квартиры в г. Москве”
Request 14	Николаева Елизавета Фотиевна	“Хочу оформить документы для получения визы, чтобы поехать в отпуск.”
Request 15	Белоусов Ладимир Твердиславович	“Беру для новой визы”
Request 16	Дроздов Епифан Германович	“Нужна справка для оформления визы, прошу сделать побыстрее”
Request 17	Зыков Адам Егорович	“Беру справку о трудоустройстве для съема жилья”
Request 18	Бобылева Вера Афанасиевна	“Нужна справка для оформления новой визы”

The manual creation of these references was assigned to a real HR employee and a comparison of the results is presented in the Table 15-17 below.

Table 15. Results for the third scenario

	Request 1	Request 2	Request 3	Request 4	Request 5	Request 6
Human HR-employee	10 min 40 sec	10 min 53 sec	11 min 8 sec	10 min 45 sec	9 min 12 sec	11 min 47 sec
RPA-bot	51 sec	48 sec	53 sec	49 sec	50 sec	47 sec

Table 16. Results for the third scenario

	Request 7	Request 8	Request 9	Request 10	Request 11	Request 12
Human HR-employee	13 min 45 sec	10 min 48 sec	8 min 18 sec	9 min 56 sec	10 min 32 sec	14 min 21 sec
RPA-bot	50 sec	54 sec	51 sec	51 sec	49 sec	50 sec

Table 17. Results for the third scenario

	Request 13	Request 14	Request 15	Request 16	Request 17	Request 18
Human HR-employee	13 min 16 sec	15 min 6 sec	14 min 56 sec	11 min 54 sec	13 min 36 sec	12 min 34 sec
RPA-bot	52 sec	50 sec	48 sec	50 sec	52 sec	51 sec

8. Conclusions

After obtaining the performance results and measuring the execution speed for all three RPA bots created during this work, it is possible for us to formulate conclusions. For each scenario, we conducted a series of tests in order to measure possible performance deviations for different user requests.

8.1 Profile search

The requests for applications were quite detailed and contained a lot of additional parameters. A person who searched for such profiles manually had to spend time on opening the browser, navigating to “hh.ru”, authentication, selecting “search candidates” menu item, filling the necessary tags on the search form, getting and validating results, and lastly transferring the results to Excel spreadsheet and a database manually (field-by-field). Alternatively, our RPA-bot performs all the specified actions in auto-mode (for example, accesses the server directly with URL with embedded tags, uses XPath technology to parse the data from the web pages, and use out-of-the-box integration capabilities to populate data into Excel spreadsheet and database simultaneously).

Based on the performed test results we can make the following conclusions:

- RPA-bot reduces the execution time to complete a vacancy search request from ~3.5 minutes to ~24 seconds per one search profile.
- RPA-bot will be more effective in case we need to process more vacancy search requests. RPA-bot works 24/7 and does not have to take a break or switch to another task in between.
- During our tests, RPA-bots did zero mistakes.
- RPA-bot uploads the data and search parameters directly to the company's server; an HR employee would have needed additional time to connect to the system and enter the data using the standard method. More integrated

informational systems the company has, the greater efficiency will be achieved with RPA technology.

8.2 Calculation of salary data

Normally, HR employee does all the calculations with the help of third-party software, like CRM or ERP tool, MS Excel or manually. Our solution allows to increase the total speed and performance of calculating dynamic salaries. The key difference is the time spent on connecting to the database and entering the necessary values and formulas into the system interface. The RPA-bot produces the final result in less than 10 seconds; unlike the manual process, the bot only asked for the required data (year, month) for which the calculation has to be made. In our tests, we have demonstrated that RPA bots, due to their integration capabilities and flexibility, can be extremely effective in optimizing processes that consist of uniform and repetitive database queries.

8.3 Employment certificate bot

In this scenario, we managed to achieve a significant acceleration of document generation with no data errors. As an example, we have started the bot 18 times, and recorded the resulting execution time. In all the cases, the bot performed the task almost ten minutes faster than a human; whereas a human needed almost 11 minutes to compose and send the document to Telegram chat to the applicant, RPA-bot handled this task in ~50 seconds. With large volumes, RPA-bot saves a significant amount of time, it gives almost 16 hours of economy for every 100 documents. It also gives a significant financial value.

In addition, the bot recorded a log of issued certificates: information about issued certificates, applicants, dates and written explanations of the reasons for the request was recorded in the company's database, which makes it possible to create different kind of reports and KPIs.

8.4 Common conclusions

Concluding the results of our work, it should be noted that we managed to show in practice the advantages of RPA-technology application for HR tasks automation. While working on this project, we have identified the areas of HR where RPA automation could significantly speed up the work processes overall. We also saw that using RPA technology reduces the number of errors to zero, it solves the problem of integration with internal and legacy systems. In addition, RPA technology significantly increases the efficiency and performance due to 24/7 continuous work without holidays, weekends, vacations, and sick leaves. What is even more important, with RPA technology we can free large amount of employees' time to perform more intellectual and decision-making work.

9. Further development

Considering the future of the work we have done, we can say that it has a great potential regarding the further optimization of performance of HR activities that we have discussed here. This program project has covered only a fraction of the activities that HR performs. Therefore, we suggest that this work has many options for further development, including deepening into the specific areas of HR activities and creating more detailed program solutions based on what we have done in this work.

For the recruitment process, for example, we may suggest the development of a digital solution for a complete multi-level approval system for the onboarding process. In this case, we can use the bot we have developed for profile searching as a basis for the future RPA-project of onboarding automatization, because the search of candidates is always one of the first and most important steps during the process of recruiting new employees. This project will also involve integration with the company's BPM or ERP systems, as well as working with the organization's database.

The future project may evaluate the usefulness of robotization for candidate approval processes, forming job offers, automatic creation of new employee's accounts in company's active directory, and so on. So, there are lots of opportunities for further development of the project in this area.

We can also add some ideas about applying AI technology together with RPA for optimizing some other HR processes. For example, we thought about extension to our work, which would imply the development of AI model for analyzing the parsed resumes and determining the most suitable candidates for the position. This project could demonstrate the effectiveness from the collaborative usage of RPA and AI, as well as demonstrate the acceleration of this business process execution in practice. Moreover, various machine learning and neural network techniques could be used for classification of the resumes collected by our bot, which could allow HR employee to select the appropriate types of candidates much faster, without reading through all of them.

The future solution could also focus on more detailed robotization of personnel management functions. In our work, we have considered only one type of requests, and made it possible to create them directly via Telegram. In the future, it is possible to expand this part of the project and make a larger RPA-solution that could create a larger number of different documents, as well as make requests within the corporate portal and forward documents for signing. Thus, we could speed up the creation of a greater variety of documents and make it possible to do all these processes via the messenger.

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11. Appendix

11.1 Main Terminology and Definitions

Main terminology, abbreviations and definitions could be found in Table 18.

Table 18. Main terminology and abbreviations used in this work

Terminology	**Explanation**
Robotic Process Automation (RPA)	Robotic Process Automation (RPA) is a technology that implies the creation of software robots (attended or unattended, or in other words that work with or without human intervention respectively) for emulating humans' actions of interacting with digital systems and software, performing repetitive and routine tasks, or integrating with 3rd-party systems (especially legacy).
HR department	HR is the organizational unit that is responsible for human resources management: finding, recruiting, hiring and training new employees for the company, as well as supporting the workforce and maintaining the harmony and wellbeing as an employee.
RPA-bot	RPA-bot is the script that emulates the sequence of human activities in the digital environment. Such bots are usually performing the tasks that do not require decision making, but take too much time and efforts from the employees. A bot can be attended (human's intervention required) or unattended (works without constant control of a human employee).
Repetitive, monotonous or rule-based tasks	Repetitive, monotonous or rule-based tasks are the tasks that require no or minimum intellectual work, take lots of time, have repetitive nature and based on some set of rules or expressions.
C&B (Compensations and Benefits)	C&B is one of the core functions of the HR department, which is focused on the management of payrolls and benefits.
BPM (Business Process Management) software	BPM software is the set of tools that helps companies to manage and automate their business processes, thus increasing the speed and quality of how their business operates. Such software

	normally includes tools for data management, forms and workflows development and execution.
ERP (Enterprise Resource Planning) software	ERP software is a set of tools that helps to optimize the day-to-day operations of core business processes, usually related to finance and supply chains.
Active Directory	Active Directory (AD) is a directory service of maintaining users and organizational units' information from Microsoft.
MSSQL Server	MSSQL is a relational database engine that uses the T-SQL syntaxis for doing various operations with data. This engine is used for handling various systems, including transaction processing, business intelligence, and data analytics.
Pix RPA	Pix RPA is a Robotic Process Automation (RPA) platform that was developed by Pix Robotics (Russia). This is a low-code platform that allows the creation, launch and administration of RPA-bots.
Telegram Messenger	Telegram Messenger is a popular messenger application, that is widely used and provides instruments for creating and administrating chat-bots.

11.2 Link to the implemented RPA-bots (GitHub)

[11] Link to the GitHub: https://github.com/romakholinov/Thesis_Project_2025

[12] Link to the web-interface for working with parsed profiles: https://k2demo.ru/Runtime/Runtime/Form/CandidateSearch.List/